

Nicholas Penha Malaya

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EDUCATION

The University of Texas at Austin, Austin, TX

- Ph.D., Engineering, 2016, Advisor: Professor Robert D. Moser
- M.S., Engineering, 2009

Georgetown University, Washington, D.C.

- B.S., Physics & Mathematics, *with honors*, 2007

RESEARCH

INTERESTS

High Performance Computing, Computational Science, Machine Learning, Deep Learning, Exascale Computing, Turbulence, Verification Validation & Uncertainty Quantification, Bayesian Inference

PROFESSIONAL PROFILE

Senior Fellow, Advanced Micro Devices (AMD). Computational scientist and technical leader in exascale computing, AI infrastructure, and AI for Science, leading software architecture, application readiness, and customer enablement for leadership-class HPC and sovereign AI systems.

SELECTED LEADERSHIP

- Technical leadership for exascale application performance and software readiness across Frontier, El Capitan, and next-generation leadership-class systems.
- Key contributor to AMD's HPC and AI-for-science strategy, spanning GPU/APU architecture, ROCm software, communication libraries, and customer workload enablement.
- Partnered with national laboratories, hyperscalers, universities, and sovereign AI organizations to translate emerging hardware capability into scientific and business impact.
- Public technical voice for high-precision scientific computing, heterogeneous open software, and the convergence of modeling, simulation, and AI.

RESEARCH EXPERIENCE

AMD

Austin, TX

Senior Fellow

2026 – present

Fellow

May 2023 – 2026

Principal Member of Technical Staff

July 2021 – 2023

Senior Member of Technical Staff

July 2019 – 2021

<i>Member of Technical Staff</i>	July 2018 – 2019
<i>Senior Engineer</i>	Dec. 2016 – May 2018

Department of Mechanical Engineering The University of Texas at Austin	Austin, TX
<i>Doctoral Candidate</i>	Jan. 2014 – Dec. 2016

Institute for Computational Engineering and Sciences The University of Texas at Austin	Austin, TX
<i>Research Engineering/Scientist Associate II</i>	Jan. 2010 – Dec. 2013

Physics Laboratory, Optical Technology Division National Institute of Standards and Technology	Gaithersburg, MD
<i>Assistant Researcher</i>	May 2005 – August 2005

PROFESSIONAL METRICS

- Google Scholar citations: 1,600+
- Google Scholar h-index: 18
- 40+ publications indexed across high-performance computing, computational science, and AI for Science
- 20+ issued and pending patents

AWARDS

- Executive Spotlight Award, June 2020
 - "For the winning LUMI EuroHPC bid"
- Next 5% Award, Q1 2020, Q3 2022, & 2025
 - "The Next 5% Award is the highest level of recognition at AMD"
- Datacenter Solution Group's Award of Excellence for work on the Frontier Exascale Proposal, 2019
- "Best Student Paper Finalist" – SC13, Denver, CO
- "Best Paper" – 2012 TACC-Intel Highly Parallel Computing Symposium
- Cockrell School of Engineering Thrust 2000 Ph.D. Fellowship
- 2007 Treado Medal, Georgetown University

MENTORSHIP

Co-ops Advised:

- Alberto Acevedo, Brett Leroux, Michael Curry (Summer 2020)
- Octavi Obiols Sales (Summer 2019)
- Artem Vysogorets, Tingting Ou, Kuanhao Jiang, Parmida Davarmanesh (Summer 2019)
- Naman Meheshwari (Fall 2018)
- Patrik Gerber, Jiajing Guan, Elvis Nunez, Kaman Phamdo (Summer 2018)
- Zhaoqi Li, Catalina Vajiac, Yu Ma, Yunkai Zhang (Summer 2017)

TEACHING EXPERIENCE

The University of Texas at Austin

CSE 380 - Tools/Techniques of Computational Science **Fall 2015, 2016**

Los Alamos National Laboratory

Computational Physics Student Summer Workshop **Summer 2012 - 2017**

REFEREED JOURNAL PUBLICATIONS

A Research Retrospective on AMD's Exascale Computing Journey Loh *et al.* ISCA '23: Proceedings of the 50th Annual International Symposium on Computer Architecture June 2023

Evaluating GPU Programming Models for the LUMI Supercomputer Markomanolis *et al.* SCFA 2022: Supercomputing Frontiers, Jul 1, 2022

SURFNet: Super-resolution of Turbulent Flows with Transfer Learning using Small Datasets Octavi Obiols-Sales, Abhinav Vishnu, **Nicholas Malaya**, Aparna Chandramowliswaran Proc. 30th International Conference on Parallel Architectures and Compilation Techniques (PACT), Aug 5, 2021

Vandermonde Wave Function Ansatz for Improved Variational Monte Carlo Alberto Acevedo, Michael Curry, Shantanu H. Joshi, Brett Leroux, **N Malaya** ACM/IEEE Supercomputing Conference 2020, November 11th

CFDNet: a deep learning-based accelerator for fluid simulations O Obiols-Sales, A Vishnu, **N Malaya**, A Chandramowliswaran, International Conference in Supercomputing (ICS) 2020, June 29th

McMahan JA, Williams BJ, Smith RC, **Malaya N.**, A Linear Regression Framework for the Verification of Bayesian Model Calibration Algorithms. ASME. J. Verif. Valid. Uncert. 2017. doi:10.1115/1.4037705.

Graham, J., Kanov, K., Yang X.I.A., Lee M.K., **Malaya, N.**, Lalescu, C.C., Burns, R., Eyink, G., Szalay, A., Moser, R.D. & Meneveau, C. “A Web Services-accessible database of turbulent channel flow and its use for testing a new integral wall model for LES.” *Journal of Turbulence* (2015)

M. Lee, **N. Malaya**, Rhys Ulerich, Robert D. Moser, “Experiences from Leadership Computing in Simulations of Turbulent Fluid Flows,” *Computing in Science & Eng.*, vol. 16, no. 5, 2014, pp. 24–31.

T. A. Oliver, **N. Malaya**, R. Ulerich, and R. D. Moser, “Estimating uncertainties in statistics computed from direct numerical simulation,” *Phys. Fluids* 26, 035101 (2014). <http://dx.doi.org/10.1063/1.4866813>

M. Lee, **N. Malaya**, and R. D. Moser, “Petascale direct numerical simulation of turbulent channel flow on up to 786k cores,” in *Proceedings of the 2013 ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*. ACM Press, 2013 (**Best Student Paper Finalist**)

N. Malaya, K. C. Estacio-Hiroms, R. H. Stogner, K. W. Schulz, P. T. Bauman, G. F. Carey, “MASA: A Library for Verification Using Manufactured and Analytical Solutions”, *Engineering with Computers*, 29(4), 487–496, 2013

N. Malaya, T. Oliver, K. C. Estacio-Hiroms, “Manufactured Solutions for the Favre-Averaged Navier-Stokes Equations with Eddy-Viscosity Turbulence Models”, *Technical Paper*, 50th AIAA ASM, 2012.

Nicholas Malaya, Karl W. Schulz, Robert D. Moser “Petascale I/O using HDF-5”, *Teragrid’10 Technical Paper*, Association for Computing Machinery, August 2, 2010.

Robert D. Moser, **Nicholas P. Malaya**, Henry Chang, et. al., “Theoretically based optimal large-eddy simulations”, *Physics of Fluids*, October 23, 2009.

BOOK

CHAPTERS

Damon McDougall, **Nicholas Malaya**, Robert D. Moser, “The Parallel C++ Statistical Library for Bayesian Inference: QUESO”, *Handbook of Uncertainty Quantification*, edited by Roger Ghanem, David Higdon, Houman Owhadi, (Springer International Publishing, 2017)

Nicholas Malaya, Robert D. Moser, Amitabh Bhattacharya. “Modeling Multi-point Correlations in Wall-Bounded Turbulence”, *Progress in Wall Turbulence: Understanding and Modeling*, edited by M. Stanislas, J. Jimenez, and I. Marusic (Springer, Netherlands, 2011), Vol. 14, pp. 183–190.

PATENTS

Deliberate conditional poison training for generative models, N Malaya, US Patent App. 16/208,384

Optimized and scalable sparse triangular linear systems on networks of accelerators, K Hamidouche, MW LeBeane, NP Malaya, JL Greathouse, US Patent App. 16/044,145

Method and system for opportunistic load balancing in neural networks using metadata, N Malaya, Y Eckert, US Patent App. 16/019,374

Configuring Computational Elements for Performing a Training Operation for a Generative Adversarial Network, NP Malaya, US Patent App. 16/009,089

Dynamic hardware selection for experts in mixture-of-experts model, N Malaya, N Jayasena, US Patent App. 15/849,633

Concurrent training of functional subnetworks of a neural network, D Yudanov, NP Malaya, US Patent App. 15/841,030

Dynamic, variable bit-width numerical precision on fpgas for machine learning tasks, NP Malaya, EH Mednick, US Patent App. 15/833,287

Artificial neural network reduction to reduce inference computation time, N Malaya, US Patent App. 15/638,993

PAPERS IN
PREPARATION

SURFNet: Super-resolution of Turbulent Flows with Transfer Learning using Small Datasets Octavi Obiols-Sales, Abhinav Vishnu, Nicholas Malaya, Aparna Chandramowlishwaran <https://arxiv.org/abs/2108.07667>

REFEREED
CONFERENCE
PUBLICATIONS

Hao Lu, Michael Matheson, Noel Chalmers, Aditya Kashi, **Nicholas Malaya**, and Feiyi Wang, “Insights from Optimizing HPL Performance on Exascale Systems: A Comparative Analysis of Panel Factorization,” Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC25), 2025.

Alan Smith, Gabriel H. Loh, Michael J. Schulte, Mike Ignatowski, Samuel Naffziger, Mike Mantor, Mark Fowler, Nathan Kalyanasundharam, Vamsi Alla, **Nicholas Malaya**, Joseph L. Greathouse, Eric Chapman, and Raja Swaminathan, “Realizing the AMD Exascale Heterogeneous Processor Vision,” Proceedings of the 51st Annual International Symposium on Computer Architecture (ISCA), Industry Product Track, 2024.

Paul T. Bauman *et al.*, including **Nicholas Malaya**, “Experiences Readyng Applications for Exascale,” Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC23), 2023. **Best Paper Finalist.**

Nicholas Malaya, Shuai Che, Joseph L. Greathouse, René van Oostrum, Michael J. Schulte, Accelerating Matrix Processing with GPUs Published in the Proceedings

of the 24th IEEE Symposium on Computer Arithmetic (ARITH 24), July, 2017

N Malaya, D McDougall, C Michoski, M Lee, CS Simmons, Experiences Porting Scientific Applications to the Intel (KNL) Xeon Phi Platform Proceedings of the Practice and Experience in Advanced Research Computing 2017 on Sustainability, Success and Impact

K. W. Schulz, R. Ulerich, **N. Malaya**, P. T. Bauman, R. Stogner, and Chris Simmons, “Early Experiences Porting Scientific Applications to the Many Integrated Core (MIC) Platform”, TACC-Intel Highly Parallel Computing Symposium, Austin, TX, April 10–11, 2012. Winner - Best Paper.

Rhys Ulerich, Kemelli C. Estacio-Hiroms, **Nicholas Malaya**, and Robert D. Moser. A transient manufactured solution for the compressible Navier–Stokes equations with a power law viscosity. In 10th World Congress on Computational Mechanics, São Paulo, Brazil, July 2012. doi: 10.5151/meceng-wccm2012-16661. (pages 108, 223, 227).

Juan Sillero, Javier Jimenez, Robert. D. Moser, **Nicholas Malaya**, “Direct Simulation of a zero-pressure gradient turbulent boundary layer up to $Re_\theta = 6650$ ”, Technical Paper, 13th European Turbulence Conference, September, 2011.

INVITED **Nicholas Malaya**, “Beware the Jabberwock: Open Software for a Heterogeneous
PRESENTATIONS HPC Future”, Salishan Conference on High-Speed Computing, April 2026.

Nicholas Malaya, “The Unreasonable Effectiveness of FP64”, HPC Smoky Mountain Conference, 2025.

Nicholas Malaya, “Realizing the AMD Exascale Heterogeneous Processor Vision”, ISCA Industry Product Track, Buenos Aires, June 2024.

Nicholas Malaya, “Decentralized Compute Infrastructure for HPC and AI”, Birds of a Feather, SC25, November 19th, 2025.

Nicholas Malaya, “Communication Libraries in HPC and AI”, Birds of a Feather, SC25, November 19th, 2025.

Nicholas Malaya, “Mixed Feelings About Mixed Precisions”, Birds of a Feather, SC25, November 18th, 2025.

Nicholas Malaya, “The EPYC CPU and INSTINCT MI250X GPUs in Frontier”, FRONTIER TRAINING WORKSHOP, Oak Ridge Leadership Computing Facility, February 15th, 2023.

Nicholas Malaya, “Performance Results at Exascale in HPC and ML”, 7th International Workshop on Extreme Scale Programming Models and Middleware, SC22, Nov 14th, 2022.

Nicholas Malaya, “The Compute Architecture of the Frontier Exascale System”, University of Delaware, October 14th, 2022.

Nicholas Malaya, “Accelerating CFD with ML/AI at AMD”, ModSim 2021, October 8th, 2021.

Nicholas Malaya, “The Frontier System: Computing at Exascale”, Northeastern University, Dec. 1st, 2021.

Nicholas Malaya, “Enabling Portable Directive-Based Programming at Exascale”, WACCPD, SC20, November 13th, 2020.

Nicholas Malaya, “Hardware and Software Co-design: Preparing and Optimizing Scientific Applications for Exascale Computing”, Institute for Applied Computational Science, Harvard University February, 21st, 2020.

Nicholas Malaya, “AMD EPYC CPUs and Radeon Instinct GPUs for HPC”, El Capitan Application Readiness Kick-off Workshop, Lawrence Livermore National Laboratory, January 14th, 2020.

Nicholas Malaya, “AMD EPYC CPUs and Radeon Instinct GPUs for HPC”, El Capitan Application Readiness Kick-off Workshop, Sandia National Laboratories, Dec. 3rd, 2019.

Nicholas Malaya, “AMD GPU Evolution”, Frontier Application Readiness Kick-off Workshop, Oak Ridge Leadership Computing Facility, Oct. 8th, 2019.

Nicholas Malaya, **Max Kiehn**, “Automated Software Testing with Machine Learning”, AMD Software Tech Forum, September 9th, 2019.

Nicholas Malaya, “Learning Systems”, Department of Computer Science, University of Chicago, November 28th, 2018.

Nicholas Malaya, “Full-Stack Open-Source Supercomputing”, OpenSuCo4 Keynote at SC18 Dallas, TX, November 12th, 2018.

Nicholas Malaya, “Technology Directions in AI”, UT McCombs School of Business, Austin, TX, October 9th, 2018.

Nicholas Malaya, “Numerical Investigation of Synthetic Dust Devils”, AMD Research, Austin, TX, July 8th, 2016.

Nicholas Malaya, “Introduction to Scientific Computation”, Dell HPC Innovation Lab, Austin, TX, September 2nd, 2016.

Nicholas Malaya, “Verification, Validation and Uncertainty Quantification in Turbulence Simulations”, Texas Advanced Computing Center, Austin, TX, May 12th, 2016.

Nicholas Malaya, “Numerical Investigation of Synthetic, Buoyancy-Induced Columnar Vortices”, Sandia National Laboratories, Albuquerque, NM, March 16th, 2016.

Nicholas Malaya, “Verification & Software Quality in Scientific Computing”, Los Alamos Computational Physics Student Summer Workshop, 2014.

Nicholas Malaya, “The Method of Manufactured Solutions”, Los Alamos Computational Physics Student Summer Workshop, 2013.

Nicholas Malaya, Christopher Simmons, “Scientific Software Engineering Best Practices”, Los Alamos Computational Physics Student Summer Workshop, 2012.

Nicholas Malaya & Robert D. Moser, “Tools and Techniques for Code Verification using Manufactured Solutions.”, SIAM Conference on Uncertainty Quantification, April 2-4, 2012, Raleigh, North Carolina

Nicholas Malaya, Karl W. Schulz, “Verification through the MASA Library”, Los Alamos Computational Physics Student Summer Workshop, 2011.

Nicholas Malaya, Rhys Ulerich, Robert Moser, “Petascale Direct Numerical Simulations of Turbulent Channel Flow”, ESP Kick-off Workshop and Project Plan Presentation, Oct. 18th, 2010.

Nicholas Malaya “Theoretically Based Optimal LES”, TFS/NRE Seminar, Department of Mechanical Engineering, Oct. 29th, 2009.

CONFERENCE Alistair Hart, **Nicholas Malaya**, *et al.* LUMI - Delivering Real-World Application
PRESENTATIONS Performance at Scale through Collaboration Cray User Group 2023

Sam Antao, **Nicholas Malaya**, Matthew McIntire, Chuck Gilbert Getting data where it’s needed: potential for CPU, GPU, accelerators, and smart networks International Workshop on Smart Networks, Data Processing and Infrastructure Units ISC 2023

N. Malaya The AMD MI250X Crusher Workshop, OLCF, 2022

N. Malaya Frontier Center of Excellence Update: AMD hardware and software ECP All-Hands, April. 16th, 2021

N. Malaya Accelerated Computing at AMD Research UT Austin ECE Speaker Panel, October 23rd, 2020

N. Malaya Preparing Scientific applications for Exascale Computing UCLA IPAM, July 15th, 2020

N. Malaya The Frontier Node Architecture Design ECP All-Hands, Feb. 6th, 2020

N. Malaya Preparing Scientific Applications for Frontier SC19 Booth Talk, Nov. 20th, 2019

N. Malaya Porting Scientific Applications using HIP SC18 Booth Talk, Nov. 13th, 2018

N. Malaya AMD Exascale Applications and Software Technologies PathForward Review Meeting, March 21st, 2018

N. Malaya, R. Stogner, R. Moser, Numerical Investigation of Synthetic Buoyancy-Induced Columnar Vortices, Bulletin of the American Physical Society 60, 2015

Nicholas Malaya MASA: A Tool for the Verification of Scientific Software, SciPy2016

N. Malaya, R. Ulerich, T. Oliver, R. Moser, Estimating Uncertainties in Statistics Computed from DNS, APS Meeting Abstracts 1, 21004, 2012

Nicholas Malaya, Karl Schulz, “Verification through the MASA Library”, DOE PSAAP Annual Review, 2011

Nicholas Malaya, Pk Yeung, “Collaborative Research: Enabling Discovery in High Reynolds Number” Turbulence via Advanced Tools for Petascale Simulation and Analysis, 2010

Nicholas Malaya, Amitabh Bhattacharya, Robert Moser, “Theoretically Based Optimal LES”, APS DFD, 2008.

TECHNICAL SKILLS

Programming Experience:

- Languages: C/C++, Fortran, Python(Num/Scipy), $\text{\LaTeX}$, ROCm/HIP/HCC, CUDA, Bash, Mathematica, Octave/Matlab
- Development Env: Linux, Emacs, Git, SVN, Buildbot, TravisCI
- Stanford Certificate in Machine Learning (Online) from Andrew Ng
- Libraries: MPI, OpenMP, Tensorflow, PyTorch, FFTW, BLAS, Lapack, Dakota, Petsc

SERVICE

Mentorship:

- Industry Mentor, RIPS IPAM UCLA, 2017, 2018, 2019, 2020
- Senior Judge, Austin Energy Regional Science Festival 2014-present

Committees:

- Vice Chair, Post-Moore Computing, SC26 Technical Program Committee
- Technical Program Committee Member, Session Chair, Programming Frameworks and System Software area at SC22

- Steering Committee Member, WACCPD SC21 and SC22
- Program Committee for Cluster 2021
- Program Committee HiPar at SC21
- Program Committee Member, SciPy 2017, 2018, 2019
- Program Committee Member, PEARC17
- Program Committee, Great Lakes Consortium for Petascale Computation (2012–2015)
- Reviewer, Blue Waters Graduate Fellowship Program (2014–2018)

Companies Advised:

- Board Member, kWh Analytics (2012-2016)

Journal Referee:

- Machine Learning and the Physical Sciences, NeurIPS, 2019-2020
- Parallel Computing
- Journal of Fluid Mechanics
- Engineering with Computers
- Journal of Computational Physics
- Neural Information Processing Systems

MEMBERSHIPS (Senior Member) ACM, SIAM

INTERESTS Marathons, Motorsport, Mountain Climbing, Wine, Go

CITIZENSHIP United States of America

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